058

UNIVERSITY OF COLOMBO, SRI LANKA

UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING

UCSC

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

Academic Year 2020/2021 – First Year Examination – Semester I – 2021

SCS1207 – Software Engineering I

TWO (2) HOURS

PART A

To be completed by the candidate

Examination Index No: ....

Important Instructions to candidates:

1. The medium of instruction and question is **English**.
2. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
3. Note that questions appear on both sides of the paper. If a page is not printed, please inform the supervisor immediately.
4. Write your index number on each and every page of the Question paper.
5. **This paper consists of two parts, Part A (Question No 1) and Part B (Question No 2, Question No 3 and Question No 4) and submit separately.**
6. Part A has one (1) question and **05** pages. Part B has three (3) questions and **10** pages.
7. Answer **ALL** questions in both Part A and Part B.
8. Any electronic device capable of storing and retrieving text including electronic dictionaries and mobile phones are **not allowed**.
9. **Non-Programmable** calculators are **allowed**.

For Examiner's use only	
Question No	**Marks**
1	
Total	

Index No: ....

Part A

Question 1

(a) Define the term **Software Engineering.**

[3 marks]

(b) Identify the type of software suitable for the description given in column (A) from the following list and write in the column (B).

Embedded software, Systems software, Artificial intelligence software, Real time software, Scientific software

[5 marks]

A	B
Software that heavily interacts with hardware and required to use the computers.	
Software that monitors, collects, formats and analyses information from an external environment and produces required information within strict time constraints	
Software that resides in read-only memory and is used to control products and systems	
Software characterized by number-crunching algorithms and used in space shuttle orbital dynamics, astronomy etc	
Software used in Pattern recognition, Expert Systems and Game playing	

Index No: ....

(c) Classify the software quality attributes given below into **current usefulness** and **potential usefulness**.

[5 marks]

Software Quality Attribute	**Classification**
Security	
Portability	
Efficiency	
Reusability	
Robustness	

(d) Give **three (3)** problems associated with the Waterfall Model.

[6 marks]

(e) Draw a diagram to show the stages of the Evolutionary Prototyping model.

[6 marks]

(f) Give **three** (3) common features of agile software development methods.

[3 marks]

Index No: ....

(g) Related to the SCRUM methodology explain the following terms.

[6 marks]

Backlog :
Sprint :
Scrum Meeting :

(h) Describe the following requirement types giving each an example.

[6 marks]

Functional requirements
Non-functional requirements
Domain requirements

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Question No	**Marks**
2	
3	
4	
Total	

Index No: ....

Part B

Question 2

(a) Different graphical models can be used in the requirement engineering and system design stage to represent the software systems.

(i). Briefly explain why several models are required to represent the system thoroughly.

[2 marks]

(ii). Briefly explain the purpose of drawing the below four (4) UML models and mention the perspective represented by each model.

[6 marks]

Use-Case Diagram
Context Diagram
Sequence Diagram
Class Diagram

(b) Briefly explain the term '**Model-driven Engineering**'.

[2 marks]

(c) Briefly explain the different views of the **4+1 View Model Architecture**.

[5 marks]

(d) Identify the best architectural pattern for the scenarios given below. Justify your Answer. [5 marks]

(I) Scenario 1:

A file server that acts as a centralized location for files needs to be designed and developed. Centrally stored and shared files can be accessed by multiple users from several locations. Servers can be replicated and may be used when the load on a system is variable.

BEST ARCHITECTURAL PATTERN:

JUSTIFICATION:

(II) Scenario 2:

A generic desktop application system needs to be developed to manage patients suffering from mental health disorders. The system builds new facilities on top of the existing system. The development team is spread across several teams and each team is responsible of for set of functionalities. Multi-level security is a high-level need to ensure patient information is stored ensuring information security.

BEST ARCHITECTURAL PATTERN:

JUSTIFICATION:

Index No: ....

Question 3

(a) Design patterns represent the best practices used by experienced object-oriented software developers.

(i). What is meant by a design pattern? Briefly explain.

[2 marks]

(ii). List down **two (2)** examples for design patterns.

[1 mark]

(iii). Briefly explain the main difference between *Design patterns* and *Architectural patterns*.

[2 mark]

(b) Implementing a software system can be done from various calipers.

(i). List down **two (2)** things that can be reused at the "*Abstract*" level.

[2 marks]

(ii). When reusing software components, one should investigate the cost of the time spent in searching for reusable software components. Justify the statement.

[3 marks]

(iii). Briefly explain the main aim with respect to the below terms.

[5 marks]

Configuration Management:

Host Target Development:

(c) Briefly explain the term '*Software Re-engineering*'—mention **two (2)** advantages of this process.

[5 marks]

Question 4

(a) Software inspections and testing is used at different stages of the Software Verification and Validation (V&V) process.

(i). List down **four (4)** different components/outputs of the software development process where software inspections can be encouraged in the V & V process.

[2 marks]

(ii). Briefly explain **three (3)** advantages of software inspections over software testing.

[3 Marks]

(iii). Suppose a software development company hired you to test an e-Commerce web-based application. Briefly explain how you would perform the below tests to test the above-mentioned system.

[4 Marks]

Stress Testing:
User Testing:

(b) Identify the possibility of applying '*White box*' or '*Black box*' testing strategies in the below-mentioned testing phase or scenario. Use a **Tick** (✓) mark to denote the applicability.

[5 Marks]

Software Phase	White Box Testing	Black Box Testing
Interface Testing		
Stress Testing		
User Testing		
Unit Testing		
SRS Check		

(c) Test-Driven Development (TDD) is a common approach used in the software development industry.

(i). Briefly explain the term 'Test-Driven Development'.

[2 Marks]

Index No: ....

(ii). Suppose you were asked to write a simple program to add two numbers. Briefly explain how you would adapt the TDD approach in coding the above program. (Note: The code is NOT required)

[4 Marks]

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